Collaborative robotics

AY 2024-2025

Implementation of a Teach and Play approach for functional movements

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Abstract

In this work, it is proposed a procedure for the implementation of the reference trajectories, based on the “Teach and Play” approach, in order to deal with trajectory realization in upper-limb rehabilitation exoskeleton. The already mentioned “Teach and Play” approach better meets therapists’ expectations but lack consistency and optimization. To overcome these limitations, force sensing during the "Teach" phase is introduced to maintain transparency and ensure precise motion recording. In the "Play" phase, the system switches to rigid control and replay with accuracy the recorded trajectory, enabling functional movements for early rehabilitation in post-stroke patients with severe motor disabilities. A drawback of this approach is the fact that the original movements are performed by human operators, that can introduce some differences from one repetition to another one. In order to find the best trajectory possible, two algorithms have been implemented. They are used to process the data obtained during the Teach phase and to have optimized trajectory for the Play phase: the first creates the mean trajectory from a set of movements and the other minimizes the jerk of the trajectories performed in the Teach phase.

1. Motivation of the project

Patients with paralyzed upper extremities due to lesions of the central nervous system or due to injuries often receive arm therapy. The goal of this therapy is to recover motor function, improve movement co-ordination, learn new motion strategies, and prevent secondary complications (i.e. muscle atrophy, osteoporosis, joint degeneration and spasticity). In addition, the recovery of the motor function allows to re-execute multiple activities in daily life. Several studies prove that arm therapy has positive effects on the rehabilitation progress of stroke patients [1]. In fact, the restoration of arm and hand functions after stroke has a major role in neurorehabilitation. In general, upper limb rehabilitation and assistance play a key role in all injuries' recovery.

In this context, exoskeletons for rehabilitation and assistance of the upper limb offer new advantages to the execution of classical therapy sessions. Their use guarantees the capacity of performing high amounts of practice, carrying out task-oriented, interactive and customizable exercises. However, to be efficient, robots must perform trajectories that accurately reproduce physiological human movements. Since the correctness of movements enhances the relearning process with the reconstruction of correct neutral path, there is the necessity of managing the trajectory-planning issue. This is important to guarantee both safety and comfort, as well as the correct execution of the movement in a rehabilitation perspective.

Since mathematical trajectories path-planning methods have problems of redundancy, poor inter-joint coordination and low human likeness, learning the movements directly from human demonstration could be a solution, especially to deal with high complexity of physiological arm movements and to obtain human-like trajectories. Among these second presented methods, the "Teach and Play" methods are the most frequently applied but have reliability issues, repeating also all the possible recording errors. In addition, results are highly dependent on the quality of the recording and require accuracy and concentration efforts from the human teacher's side. To prevent the previous problems, it's possible to implement algorithms that process the data coming from the recording phase and reduce the influence of differences among different recordings by calculating the mean trajectory or optimizing the movements by calculating the minimum jerk trajectory.

2. Objective of the work

In this work, a "Teach and Play" path-planning procedure is proposed to deal with rehabilitation of patients with mild-severe impairments in upper-limb. This approach consists in recording a demonstration of a desired movement performed by an operator wearing the exoskeleton, and then simply recreating it to the user.

This work is aimed at subjects with acute post-stroke conditions who have severe disability for early rehabilitation setting. i.e., Muscle Research Council (MRC) scale for both Biceps and Triceps muscles is lower than 2 (i.e., no active movements against gravity). It is required to implement a Teach and Play approach for a functional movement and to develop "flexible" trajectories, based on the human-like movement. During the Teach phase, the exoskeleton acts transparently, which means that an operator (i.e. a therapist) can perform the desired movement without perceiving any resistance. Transparency is obtained by minimizing the influence of the exoskeleton on the movement by assisting the therapist's movement with some properly generated torques, so that the system can learn the movement. The trajectory is registered thanks to the use of some position sensors placed on the exoskeleton. Force sensors are used to sense the therapist intention of movement and to implement the exo's transparency.

During the Play phase, the exoskeleton switches to “rigid”. The trajectory registered is elaborated and the optimized trajectory is reproduced in order to simulate the learnt movement. The system guarantees high precision through compensation of possible external noise, keeping a fluid and natural movement and avoiding abrupt changes of force or velocity. This is a crucial phase for assisting the patient in the execution of functional movements while promoting relearning of motor skills. This approach reduces the need of complex mathematical models, making the job simpler and more intuitive for the therapist as well.

3. Requirements

3.1 Functional requirements

Functional requirements are features that allow the system to accomplish the task for which it is intended; in the case of the current project the main objective is to reproduce a trajectory, previously recorded. In order to do that, the main functional requirement is the capacity of the system to reproduce the recorded trajectory as precisely as possible in different situations. To test this requirement, it’s possible to evaluate the error between the positions recorded during the Teach phase and the ones measured during the Play phase.

Another essential feature in the rehabilitation process is the possibility for the therapist to perform different types of movements based on the conditions of the patient and also on the rehabilitation stage; this requirement can be tested through the reproduction of different exercises (different ROM) by different people.

Ease of operating is important for the systems intended to be used by non-skilled operators: the complete procedure should be fast and should not require a large number of operators to be performed, in order to be compatible with rehabilitation session timing and resources. This requirement can be tested by timing the entire procedure and evaluating the number of operators necessary to perform the task.

3.2 Non-functional requirements

Non-functional requirements are important because they enhance users experience; in the case of this project, the first non-functional requirement analyzed is the presence of a safety button, implemented to stop the reproduction of the movement if something unexpected happens; the validation can be done by trying several times the code until the button doesn’t work as expected.

Two more requirements are: the possibility to easily re-do the recording if the movement doesn’t satisfy the therapist and the possibility to reproduce the movement more than once when the exoskeleton is worn by the patient, meaning that the Teach phase is necessary only when a different type of movement is required. It’s possible to validate both these requirements by testing the algorithm of the software several times until the right functioning.

The last non-functional requirement is the transparency of the motor during the Teach phase; this is very important because the recorded movement can be more natural if the therapist doesn’t feel the influence of the exoskeleton. The transparency is influenced by a factor, called “gain”, that can be set by trying different values and choosing the one that offered the best feeling.

4. Materials and methods

4.1 Hardware and software

The implemented device is a rigid portable upper arm exoskeleton designed for rehabilitation purposes. The objective of the system is the partial or total recovery of motor control and capabilities in the flexion/extension of the elbow in post-stroke patients during the early stages of rehabilitation. The target body section of the device is the upper limb, in particular the shoulder and the elbow joints. The human shoulder joint, a ball-and-socket joint with 3 DoFs, is recreated as a 2 DoFs joint by means of two rotational joints; one rotational joint has a horizontal axis of rotation and permits the flexion/extension of the shoulder, the other has a vertical axis of rotation and permits the internal/external rotation of the shoulder. The human elbow joint is a hinge and also in this case it is recreated by a rotational joint. The only active joint is the elbow, that is activated by an electrical actuator; while the shoulder is completely passive and can support the shoulder by means of a spring.

The exoskeleton (EduExo Pro [2]) consists of a metal structure that allows securing the robotic arm to the human arm to ensure proper alignment between the anatomical and robotic axes. This aspect is crucial, as misalignment can lead to incorrect movement execution, compromising the therapeutic effect or even causing joint damage. The structure is connected to the user's torso by means of a jacket. The motor is an electric servo motor with gears and control electronics placed in the motor housing together. The system is equipped with different sensors, in this section only the ones used in the development of the device will be listed. The first one is an integrated angle sensor, that permits the accurate replication of the movement during the Play phase by recording and replicating the positions executed by the therapist. Other electronic components are the force sensor and its amplifier, that permit to measure the interaction force between the exoskeleton and the human user; the strain gauges of the force sensor change their electric resistances when deformed, permitting the detection of the applied load. The microcontroller used in the device is an Arduino Nano 33 IoT, which is a programmable electronic device that acts as a control unit. To simplify the connection of all the components to the microcontroller, a Printed Circuit Board (PCB) is used. The exoskeleton's commands are directly controlled by the user via buttons mounted on the device, which interacts with the PCB with a direct connection. In conclusion a set of six AA batteries into the battery holder is necessary to make to exoskeleton perform the movement.

From the software point of view, the exoskeleton functioning is composed of different sequential steps, into which the user could “navigate” thanks to the button. In fact, by keeping pressed the button for two seconds, both the Teach and Play phases start and could be ended; instead, by pressing the component for different time-period at the end of both phases, it is possible to re-do or keep the recorded acquisitions (for 5 and 2 seconds after the Teach phase respectively) and to start a new movement acquisition or repeat the old one (for 5 and 2 seconds after the Play phase respectively). In conclusion, the pressing of the button for a time-period lower than two seconds during the Play phase permits the emergency stop.

4.2 Kinematic and dynamic model

Necessary step of exoskeleton design consists of the definition of its kinematic and dynamic models. In fact, the definition of the procedure thanks to which the robot moves and about the relations between forces/torques and the motion is a fundamental part of the design of the collaborative robot. Different modelling is required for the different Teach and Play phases.

Regarding the Teach phase, there isn’t a proper kinematic model because the joint’s angle is directly measured by the position sensor. The most interesting aspect of this phase is the dynamic model that permits to obtain the transparent behaviour of the motor; the theoretical model that should be implemented is schemed in Figure 1. The total gravitational force of the system (forearm + “forearm” of the exo) can be obtained by knowing the mass of the forearm from anthropometrics tables and the mass of the forearm of the robot from the datasheet. Then, if the sensed force (F_{sens}) by the force sensor is different from the

normal component of the gravitational force ($P_{\perp}$), the therapist have started to perform a movement; in order to have transparency the motor should react to this force by applying a torque (τ_{mot}) such that F_{sens} is null. This procedure should be performed at every joint's angle. This is the model that should be implemented to obtain transparency from a theoretical point of view. However, in the analysed system the transparency is obtained by means of a gain factor, that updates the joint's angle based on the force sensed, because the available instruments couldn't guarantee analytical results.

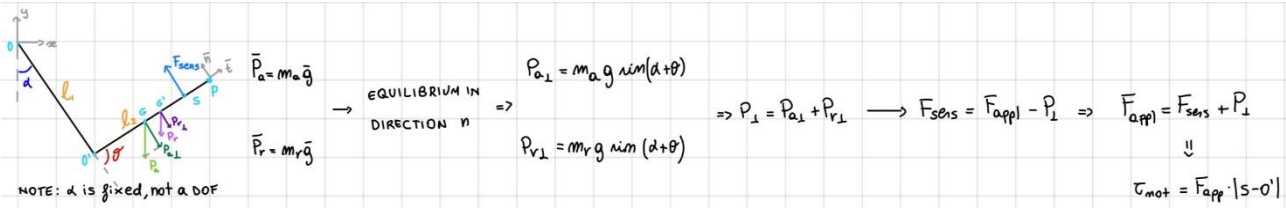

Figure 1. Dynamic model of the theoretical transparency (Teach phase)

During the Play phase, the end-effector position is directly dependant from the joint's angle and it can be obtained thanks to a Forward Kinematic Problem (FK PB); in fact the exo is asked to follow point by point the positions recorded during the Teach phase and the coordinates of the end-effector can be calculated following the relationships shown in Figure 2; from the reported equations is clear the non-linear relationship between the joint's and the end-effector's coordinates and their derivatives, but also the influence of the shoulder angular position (that is not a controlled DoF) in the end-effector positions determinations.

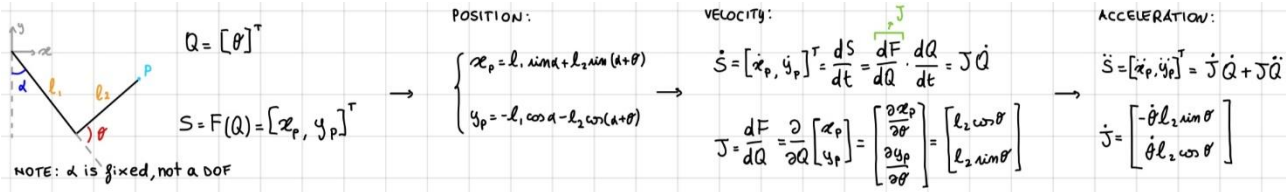

Figure 2. Kinematics relationships between joint and end-effector coordinates (Play phase)

From a dynamic point of view an Inverse Dynamics Problem (ID PB) is solved: indeed, knowing the target positions of the elbow joint (that are known from the recordings performed during the Teach phase) the exoskeleton controller is able to compute the rotational joint's torques necessary to follow the recorded trajectory.

During the Play phase both the presented kinematic and dynamic models require a reference trajectory, which is defined thanks to a Teach and Play approach. This strategy permits to define a reference trajectory without the need of any mathematical models of human behavior, with the possibility to "adapt" to unstructured environments and to preserve the anthropometric configuration (and constraints) of the user. In the presented study, the recording of a specific task's demonstration is integrated with a signal filtering, a meaningful via-points identification, an averaging of the correspondent angular values (possible thanks to different recordings of the same task) and with the computation of a minimum-jerk trajectory. In fact, only after having mediated the recorded trajectory on multiple performed trials and after having verified that the computed joint's velocity profile is a "single peak" one – bell-shaped speed profiles are typical of human-like movements – the Play phase is performed.

4.3 Control design

Hardware level of compliance

The hardware-level layer in robotics plays a critical role in achieving compliant behaviour for collaborative compliant control, particularly in applications aimed at supporting human movement as the one of the

presented study. It is possible to use different control approaches to detect user's intention of movement, while promoting transparency and compliant motion. This compliant motion is obtained thanks to the integration between actuation, transmission and sensor technologies that aims to ensure a relationship between voluntary human activity (muscle-generated torques) and robot assistance (actuator-generated torques). As a consequence of this interaction, present also in the considered exoskeleton, the total torque at the elbow joint is the sum of the torque generated by the muscles and the torque provided by the exoskeleton actuators.

Considering the hardware components described in Paragraph 4.1, the hardware control strategy selected is the one that allows a geared motor to interact with force sensors to measure the interaction force between the human user and the robot. This approach ensures precise monitoring and responsive adjustments, enhancing the device's performance and user experience.

Low-level control solution

A fundamental part of the collaborative robot design is the definition of the low-level control solutions implemented: they will be different for the “Teach” and the “Play” phases and they will be presented in this subsection.

Before talking about the effectively realised low-level control strategies, an important consideration related to the exo’s back-drivability has to be performed. The used exoskeleton is classifiable as “back-drivable”: indeed, it has the mechanical property of being able to relatively easily invert its motion when a direct power flow from the environment to the motor is present. In fact, the developed exo’s control is thought to permit the inversion of the motion while the therapist exerts a sufficient force on the end-effector.

Regarding the effective low-level controls implemented, their first important characteristic is their definition in the joints’ space: indeed, all the actors that enter in the control scheme work with angular quantities and do not refer to the end-effector Cartesian space.

Going more in detail with the description of the low-level control solutions implemented, the one developed to assure the objectives of the “Teach” phase is an “admittance control”; instead, while considering the “Play” phase a special explicit impedance “control in position” strategy is developed. Their schematic representations are presented in Figure 3 and Figure 4.

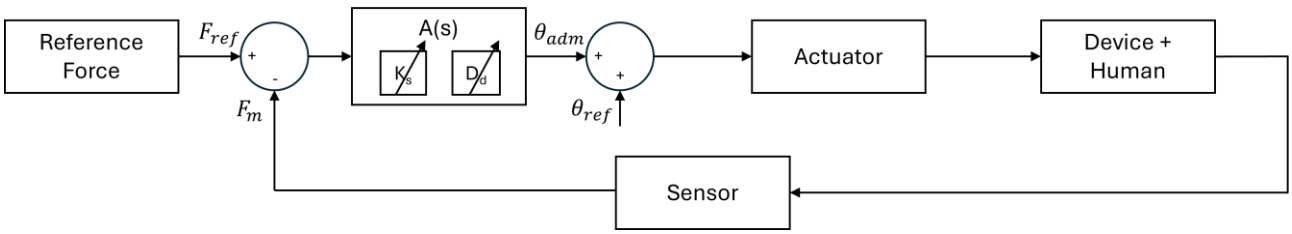

Figure 3. Admittance control diagram (Teach phase)

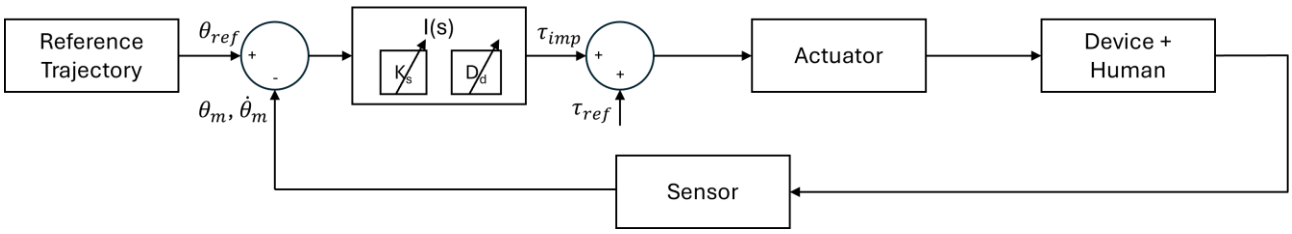

Figure 4. Impedance control – “control in position” – diagram (Play phase)

The implemented low-level control solution for the “Teach” phase is an “admittance control”, or a better said “position-based impedance control”: indeed, a measure of the forces applied on the end-effector from

the therapist is performed, and a comparison between these forces with the "reference" one (F_{ref}) – relative to the "zero-force" condition – permits to obtain the angular positions during the "Teach" phase. Through sensing the forces at the end-effector (F_m), like presented in the scheme in Figure 3, the exo's control system is developed with the ability of computing the correspondent joint's angular positions (θ_{adm}) through an "admittance controller" consisting in a "gain" factor. This "gain" factor allows to extract an angular position directly correlated to the force gradient (w.r.t. the "zero-force" reference condition) applied at the end-effector: higher forces differentials will generate highly different positions w.r.t. the position of the exo at the start of the force application. In addition, the constant comparison between the computed position with the limit reference ones (θ_{ref}, which are θ_{max} and θ_{min}) permits to obtain the wanted transparent behaviour of the exo together with a recorded trajectory always physiologically acceptable.

On the other hand, the "Play" phase low-level solution is a special impedance control called "control in position" (Figure 4). In fact, during this phase a continuous reading of the real-time angular position allows a constant comparison with the recorded reference trajectory correspondent position, and both a real-time correction – if the actual angular value is different from the already defined expected value – and definition of (new) exo's angle. Thus, the control solution developed for the "Play" phase is similar to an impedance based one, but it performs a constant comparison between reference and actual angular positions and succeeds in its definition without any "impedance controllers": this is the reason why it is called a "control in position" strategy.

The implemented low-level control strategies for the "Teach" and "Play" phases are characterized by high admittance (to guarantee a high transparency) and high stiffness/impedance (to permit to severely impaired patients to perform the recorded task with low trajectory errors) respectively. In addition, a sort of "admittance control testing" is performed to determine the best value of the "gain" parameter: this test procedure and its results are presented in the "Results" chapter.

High-level control solution

After the description of the low-level control strategies, the high-level control solutions implemented in the exoskeleton have to be presented. Also in this case a distinction between "Teach" and "Play" phases is present, and thus two different high-level control system will be presented.

Considering the residual abilities of the patient, described in Paragraph 2, the control solution implemented for the "Play" phase is thought to permit to replicate the exact pre-recorded motion (so a predefined trajectory) any times required and without taking into account any possible actions of the impaired user (no slacking problem).

In addition, since the exo must be worn also by a therapist during the "Teach" phase, the control solution requires the "transparency" of the robot during this phase: indeed, in order to record a movement's dynamics as close as possible to the real one, the medical professional has to sense the lowest possible resistance to his movements.

For these presented reasons, the exo is implemented with an "Active controller" when the "Teach" phase is considered, and with a "Passive controller" for the "Play" phase. In fact, during the recording of the task performed by therapist, the exo's high-level control solution consists in the implementation of a "transparent" behaviour through the action of the active motor and of the measuring of the force: when the force applied by the therapist is sensed at the level of the end-effector's strain gauge sensor, the motor moves the exo in the direction identified by the sign of the force. A positive force on the end-effector' handcuff means the therapist's will to move the forearm upward, vice versa a negative force. Thus, this "Active" controller is an "admittance-based assistance" one: indeed a "zero-force" reference value is defined,

and when the measured force deviates from this reference the system produced a gradient of forces (that usually vary proportional to the sensed force) that allows to implement the “transparent” behaviour of the exo.

Instead, during the “Play” phase a “Passive controller” is implemented: the exoskeleton controller does not take into account the user’s intentions, but it’s thought with a “stiff” behaviour and high forces that allow to the hardware to follow the predefined and pre-recorded trajectory with the lowest possible error. Thus, all the forces and actions are performed by the exoskeleton uniquely, the desired trajectory is created recording the robot joint angles during the “Teach” phase and in the “Play” phase the reference trajectory is exactly replayed with the only possibility to drastically stop it with an emergency button (thanks to the therapist action). In addition, the reference trajectory is obtained as the resulting trajectory characterized by the minimum jerk and deriving from the averaged positions’ sequences of three different trials performed for the same task: this guarantees the high correctness and smoothness of the replayed motion.

Intention detection strategy

Robots often employ different interaction modalities with humans, depending on the specific context of their use. For instance, they may assist users in following a predefined trajectory, provide support such as weight compensation, or adapt their actions based on users' intentions. In the context described, an intention-detection strategy (IDS) is utilized during the "Teach phase", where the robot operates transparently, aligning its movements with those of the therapist. On the other hand, during the "Play phase", the robot replays the movements rigidly, without accounting for the user's intentions. Focusing on the "Teach Phase", the intention detection strategy is categorized according to a Four-Level classification scheme (Table 1):

Level 1: Source of input signal	Level 2: Physiological phenomena of IDS	Level 3: Measured signal	Level 4: Sensors used
Wrist	Isometric force	Exerted force	Force sensor at wrist level

Table 1. Four-Level classification scheme of intention detection strategy (Teach phase)

In this scenario, the transparent mode is achieved by measuring the force exerted at the wrist. The recorded force is then used to adjust the exoskeleton's position dynamically through the gain coefficient seen in the previous chapter, ensuring the smoothness of the movement. This mechanism is activated only when the force exceeded a predefined threshold, preventing the exoskeleton from responding to small, involuntary forces applied to the sensor.

4.4 Protocol for testing and validation

This section outlines the protocol used to test individual components and the entire system. The process began with testing the individual components, starting with the button to verify its functionality. When pressed, the button is expected to record a "Low Voltage" value. To confirm this behaviour, the button was tested by pressing it for two seconds to observe if the value consistently remained “Low”. Minor oscillations or occasional high values were noted during this process.

Next, the motor's position sensor was calibrated. The exoskeleton was moved to two known positions (0° and 90°), and the corresponding tension values read by the sensor were recorded. The average of these values was calculated to establish a calibration baseline, enabling other recorded positions to be mapped accurately.

Additionally, the difference between the input positions provided to the motor and the actual measured positions was evaluated. Discrepancies, potentially caused by internal friction or suboptimal calibration, were identified. To test this, the motor was instructed to move to predefined positions, and the actual positions

were recorded. These tests also included movements performed by different subjects at various speeds to account for variability. In this case, each subject performed the same movement at three different velocities and for three trials. Each movement was recorded and then re-played to try to understand how the angles' difference changes with different velocities.

The force sensor, critical for ensuring transparency during the teaching phase, was also tested. The goal was to determine an appropriate gain coefficient that would facilitate smooth, transparent movements. This was achieved through a trial-and-error approach, relying on human feedback regarding the resistance experienced during operation.

Regarding the overall system, several key features were tested: the ability to re-do acquisitions due to errors, the option to repeat movements as desired by the user, the use of a safety button during the play Phase, and the possibility to perform different types of movements, with different RoMs. These functionalities were assessed by repeatedly running the algorithm.

Eventually, the system was validated through multiple Teach and Play phases, involving different subjects to simulate a clinical setting with both therapists and users present. The total time required to execute the algorithm was measured to ensure compatibility with real-world scenarios.

5. Results

The first component tested was the button; the test procedure was repeated three times in order to have consistency in the results. In all the three trials the values obtained were always a "Low Voltage" state, this means that in the tested conditions oscillations in the voltage state weren't present. However, it couldn't be excluded that some oscillations may happen and in order to guarantee its right functioning also in the complete system a conservative approach was used in the codes. The state of the button is acquired for a certain time and the button is considered pressed only if the button state is "Low" for 90% of the time.

As mentioned in the previous section, one of the first steps was the calibration of the position sensor, needed to map the other positions accurately; in fact, even if the position was fixed, the tension value changed. To perform the calibration, the mean tension value was taken. The mean value for 0° was 869 mV, while for 90° it was 463 mV. The calibration was quite fast to be performed (it required less than a minute to be completed). Indeed, with the implemented algorithm, the tension values were recorded for a period of time chosen by the user by pressing the button. The algorithm automatically computed the means, and the two values of tension corresponding to the two known positions were obtained in about a minute.

Once the position sensor was calibrated, the difference between some input positions (given as predefined angles) and the measured ones were studied. In Figure 5 it is possible to see the mean difference between the expected angles and the measured ones on the five trials performed. Considering all the trials, the mean deviation was $(10.5 \pm 2.5)°$.

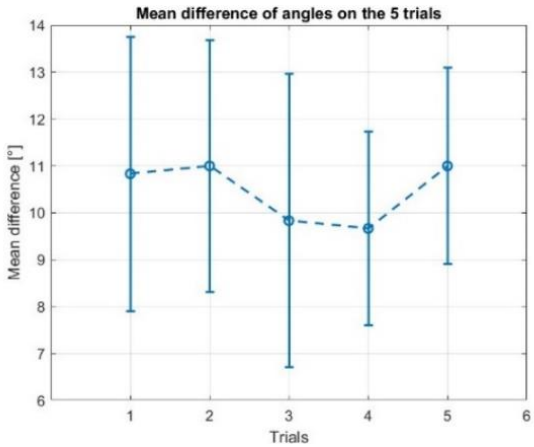

Figure 5. Mean differences between predefined input and measured angles (on five trials)

Eventually, the influence of the speed of the movement was tested involving two different subjects. In Figure 6 the fast movements recorded and replayed by the two subjects have been plotted. The same has been done considering the other two velocities, obtaining similar graphs.

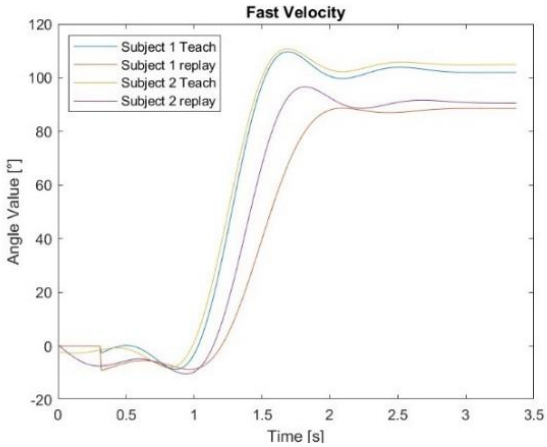

Figure 6. Recorded trajectories during both Teach and Play phases during fast movement

As it is possible to notice, there are relevant differences between the recorded ("Teach") and played ("Replay") movements. The mean difference, together with the standard deviation for the different speeds are shown in Figure 7, considering the two subjects tested.

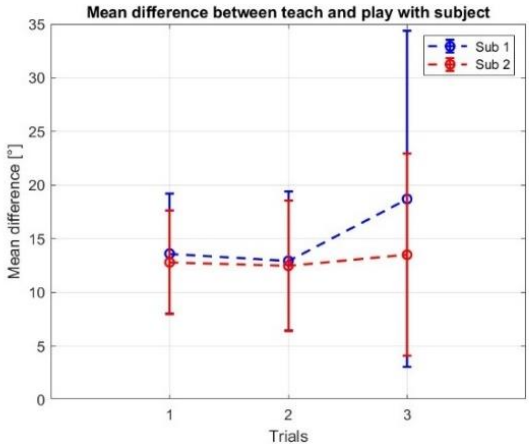

Figure 7. Mean differences and STD between measured values during Teach and Play phase and for three velocities

Subject 1 exhibited a very high variability especially in the fast movement. The mean differences considering the three speeds have been computed averaging the two subjects. The results are reported in Table 2:

Slow speed (Trial 1)	Medium Speed (Trial 2)	Fast Speed (Trial 3)
$(13.2 \pm 5.3)°$	$(12.7 \pm 6.3)°$	$(15.9 \pm 13)°$

Table 2. Mean differences between Teach and Play trajectories during movements at different speed

The results show that the variability increases with the increase in speed. These is suggested by the increase in the standard deviation, meaning that the faster the movement, the less precise it is.

Regarding the definition of the reference trajectory, a Teach and Play approach together with an averaging on different trials of the same task and a jerk minimization was considered. After having reported the results related to the calibration and to the precision of the exoskeleton, the reportage of the main outcomes regarding the procedure used to define the reference trajectory is necessary.

As presented also in the previous chapters, the implemented procedure consisted of the acquisition of the positions reached by the exoskeleton during the Teach phase and in three different trials of a 0°-100° elbow flexion. After the elaboration of the recorded signals in Excel, in Matlab software their filtering with a 6th order Butterworth filter was performed and the identification of signals' meaningful via-points (the first peak angular value above 90° of each recording) permitted their correct alignment. Successively, the computation of the "mean trajectory" was performed by averaging the three aligned positions vectors. The dis-aligned signals and "mean trajectory" are reported in Figure 8.

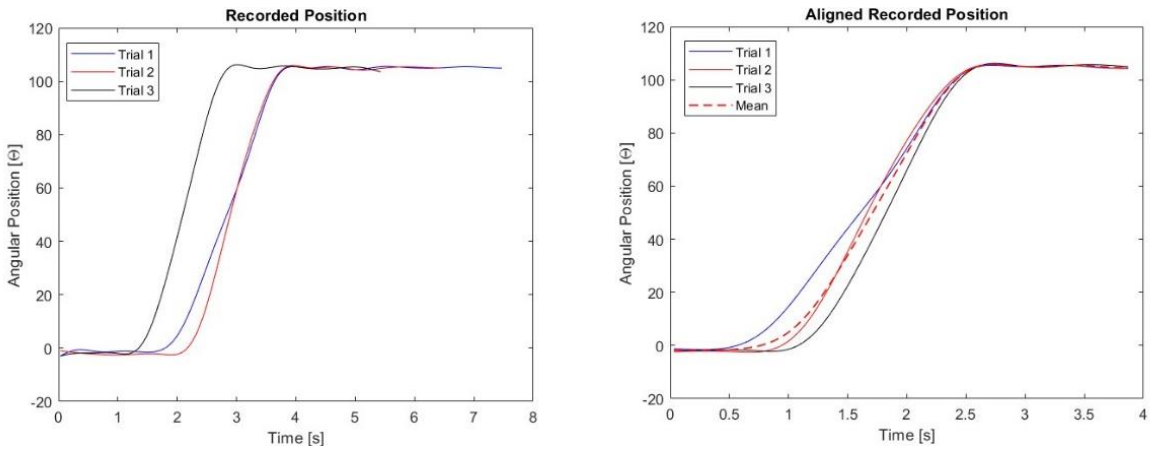

Figure 8. Recorded trajectories before (left) and after (right) alignment and averaging

Once the "mean trajectory" was computed, the main objective was the computation of reference trajectory that resemble as much as possible a human-like movement. Thus, the minimum jerk condition was required: thanks to the *"minjerkpolytraj()"* Matlab function it was possible to compute both the minimum jerk vector and the minimum jerked trajectory from the "mean trajectory". To verify the quality of the simple "mean trajectory", but also to verify that the obtained minimum jerked trajectory was the optimal one, some simple error metrics were computed. In fact, having performed three times the 3-trials acquisition – and thus having obtained three different minimum jerked trajectories – for each of these three trials acquisition the mean absolute error (MAE) and the root mean square error (RMSE) between the "mean trajectory" and the correspondent minimum jerked one were computed. Since their values are considerably small with respect to the angular values recorded during the motion (0.49° ± 0.04° for the MAE, 0.74° ± 0.03° for the RMSE), it is possible to conclude that the differences between mean trajectory and minimum jerked one are minimal for these simple recorded task: this testified that even a simple position averaging is sufficient to obtain a good human-like movement. In addition, to justify the choice of the minimum jerked trajectory, the computation of the normalized jerk score (NJS) index before (so on the mean trajectory) and after the minimization of the jerk was performed: an average NJS reduction of (0.08 ± 0.11)% testifies that the

minimum jerked trajectory is effectively the optimal one in terms of smoothness, even if the simple mean trajectory seems to be of good quality too. It is important to highlight that for the computation of the average NJS reduction only the last two 3-trials acquisitions were considered, due to some erroneous and noisy peaks in the jerk function time series of the first three recordings.

Finally, the optimality of the computed minimum jerked trajectory was evaluated also by looking at the velocity profile obtained with its differentiation: a typical bell-shaped speed profile (Figure 9) testifies the ability of this trajectory in recreating a human-like movement.

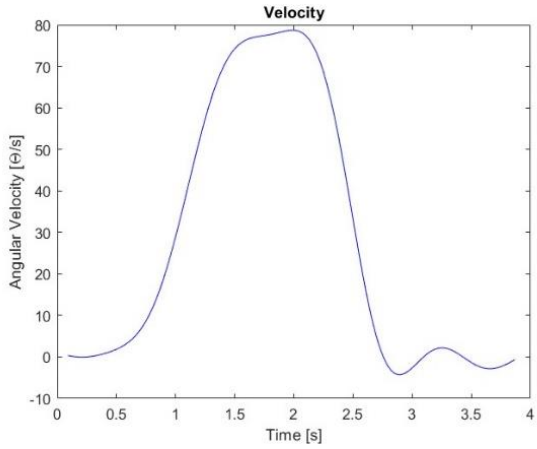

Figure 9. Velocity profile obtained from the minimum jerked trajectory

To guarantee the right transparency of the robot, the gain factor was set after different trials of the Teach phase performed by different subjects. The chosen value, 0.5, is the value for which the subjects felt the movement more natural.

In order to test the right functioning of the possibility to re-acquire a new recording of the Teach phase, the part of the code related to this activity was isolated and tested alone. As a starting point it was tested while the exoskeleton was not worn by a subject and then when the operation satisfied the wanted features, the code was tested on the exoskeleton worn by a subject. The same procedure was followed for the test of the possibility to replicate several times the recorded movement without the repetition of the Teach phase, and also for the possibilities to stop the Play phase by pressing the button in case of emergency or to record different movements. Once all these functionalities have been assessed separately, they were tested also in the complete code in order to guarantee the right communication with the other phases of the process.

Lastly, it was verified that the time required for the execution of the complete procedure was compatible with real-world scenarios, the times required for three repetitions of the Teach phase of a standard movement, the processing of the data acquired in order to retrieve the minimum jerk and mean trajectory and then three replications of the acquired movement were recorded. Two trials were done, the mean times required for the three phases are, respectively: (93 ± 5.7)s, (170 ± 42.4)s and (63 ± 5.7)s. The longest part of the procedure was the processing of the data, that had to be done by a skilled operator and the duration is highly dependent on his or her habit to perform this operation; therefore the system in the actual configuration isn't really suitable for an extensive use from everyone, but a certain knowledge is required, and further development is necessary to make the system accessible from everyone. During the processing of the data, there is time for the operators to transfer the exoskeleton from the therapist to the patient. The total time for the complete procedure was 327.5±41.7s, corresponding to 5:27 minutes; a time compatible with a real session, considering that in a single session more than one type of movement would be done. The number of operators required to help the therapist wearing the exoskeleton was only one, while in order to put it on the patient two operators were required.

6. Discussion and conclusion

In conclusion, a brief recap of the results obtained, particularly with respect to the initial objectives and requirements, is necessary.

In the presented study the implementation of a Teach and Play strategy is correctly performed: the exoskeleton could reproduce any possible recorded movement. Nevertheless, some considerable errors between the angular values of trajectories of Teach and Play phase have been found, especially when considering high speed acquisitions.

From the results, the developed control solution allows to perform different movements and with a certain "ease of operation": different operators could correctly perform the calibration and the usage phase of the device in the typically timings related to these phases.

The implementation of a safety button is correctly performed: indeed, the component is able to work with a precision higher than 90% even if some voltage oscillations are present during its functioning. Also the "usability" and "accessibility" of the exoskeleton are tested and assessed by verifying the possibility to discard a recorded acquisition or to replay a chosen reference trajectory.

Finally, also the "transparent" behavior during the Teach phase is correctly implemented by manually identifying a suitable gain factor able to reduce the exoskeleton disturbances during the recording.

In conclusion, the main objectives are reached, but some limitations could be highlighted and could be the starting point for future developments. In fact, even if the minimum jerked reference trajectory is correctly estimated, the procedure with which the recordings are extracted and elaborated is manually, extremely time-consuming and prone to human-errors. Moreover, for how the signals are aligned in the algorithm of jerk minimization, it's only possible to perform simple movements (no consecutive flexions/extensions). Thus, a possible future work is the automatization of this procedure, together with the implementation of a Machine Learning model able to learn from the input recorded trajectories and to autonomously compute the best human-like reference trajectory (also without the need of a constant cable connection).

Additionally, since the determination of the gain factor related to the transparent behavior of the exoskeleton is manually determined on the basis of how the therapist feels the resistance provided by the device, a better definition of the transparency should be performed in future work. This will require both a more scientific based method to determine the gain factor and also a more rigorous testing phase of the realized component.

Another possible implementation is the development of a safety detection strategy based on the force measured by the forearm hand-cuff sensor: indeed, even if the safety button correctly works, a more direct method to stop the movement should consist in the therapist grabbing the arm and exerting a certain "anomalous" force to stop the motion.

Finally, also the correct determination of the limit recording time, in different contexts and with different acquired movement, should surely be a fundamental step for future works.

7. References

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